



Sustainable Development Verified Impact Standard

ANGOLA CARACULO PV PLANT



Document Prepared by

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Project Title	Angola Caraculo PV Plant
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Project Location	Angola, Namibe province, Caraculo village
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Project Lifetime	22-March-2023 – 21-March-2033 (10 years period)
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Expected Future Assessment Schedule	<i>Where known, provide the expected schedule for initial verification, or, if using project design evaluation, validation or project implementation evaluation</i>

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1 SUMMARY OF SDG CONTRIBUTIONS

Table 1: Summary of Project SDG Contributions

Row number	Estimated Project Contribution by the End of Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
<i>Sequential row number</i>	<i>Brief description of the quantifiable impact of the project's activities related to the SDG indicator</i>	<i>SDG target number</i>	<i>SDG indicator number, where directly related to an indicator, and exact text of indicator</i>	<i>Indicate the project's net expected contribution to the indicator</i>		
1)	The project will supply approximately 25 MWp of clean electricity in Angola Southern Region	7.1	7.1.1. Proportion of population with access to electricity	Increase	3.3	Label
2)	The scope of Caraculo project is to ensure a reliable production of electrical renewable power to be dispatched into the Southern Transmission Grid of the Country, which currently relies mainly on diesel thermal plants for its supply	7.2	7.2.1. Renewable energy share in the total final energy consumption	Increase	4.3	Label
3)	The project will generate a local demand for labor in the civil construction sector,	8.5	8.5.1: Average hourly earnings of employees, by sex, age,	Increase	3.3	Label

	contributing to temporarily mitigate the unemployment levels.		occupation and persons with disabilities.			
4)	With the creation of this powerplant, there will be an encouraging growth in the economy of the people living in Caraculo,	8.3	8.3.1. Proportion of informal employment in total employment, by sector and sex	Increase	3.3	Claim
5)	With the implementation of the project, there will be an increase in the production of energy from renewable source and less dependence on conventional sources, increasing reliability and, therefore, improving the local energy supply.	13.2	13.2.2. Total greenhouse gas emission per year	Decrease	4.3	Label

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

The energy sector is currently facing one of the most pressing challenges of our time. While global demand for energy continues to rise due to population growth, economic development, and urbanization rates, particularly in emerging economies, the energy supply sector remains the largest contributor to global greenhouse gas emissions. It is imperative that we adopt sustainable energy practices, in order to reduce CO2 emissions and design a sustainable energy future that changes the way we produce, deliver and consume energy.

Caraculo Project is an active participant in this transition to a greener model. It aims to address major environmental challenges while expanding economic opportunities and guaranteeing greater prosperity. In terms of SDGs contributions, the project addresses in particular the following objectives:

- **SDG 7 affordable clean energy for all**

In line with this objective, the Angola government has launched a long-term plan for the energy sector known as "Angola Energy 2025". The plan aims to provide access to basic energy services to the population and increase the supply energy mix of the country. With current power demand at approximately 5 GW and an annual growth rate of 12.5%, the overall system is expected to reach over 7 GW by 2025. To meet this growing demand, the plan involves the implementation of a photovoltaic field with a total capacity of up to 50 MWp, which will afford easy energy access to a large number of populations. This project activity represent the first phase of 25 MWp that will be followed by a second or more phases, up to other 25 MWp as soon as the electricity load of the grid increases and RNT requires the additional photovoltaic plant capacity.

- **SDG 8 Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all**

Furthermore, the construction and improvement of energy infrastructures are essential for the development of industries in the country. Renewable energy offers long-term and durable energy security, representing an investment in energy supply that contributes to both environmental needs and economic growth. The project is also contributing to the transfer of knowledge from

industrialized countries by involving local workers in the operational phase and implementing high-qualified technology. Additionally, the reduction of the country's dependency on fossil fuels is increasing the competitiveness of local industries, indirectly creating job opportunities in sustainable industry sectors.

- **SDG 10 reduce inequalities within and among the countries**

The lack of sufficient access to energy is a significant contributor to inequality within countries. In Angola, almost half of the population, particularly in rural areas, experiences "energy poverty". This condition is a significant contributor to inequality, with energy access being crucial for the achievement of a wide range of development indicators, such as health, education, food security, and gender equality. Energy access also has a social impact as it contributes to spreading information and increasing security through improved lighting facilities.

- **SDG 13 Take an urgent action to combat climate change and its impacts**

Lastly, the photovoltaic plant implemented in the Caraculo Project contest aims to reduce the actual diesel consumption and lower the CO₂ emissions in the Southern Region of the country. The initiative aims to address the steady increase of peak production in the period 2017-2022 by reducing diesel consumption for electricity generation. The installation of the first 25 MWp PV Plant would allow a reduction of diesel consumption for electricity generation of around 18,000 m³/year, in comparison to the year of 2017, where the diesel consumption of the main thermal plants of the provinces of Huila and Namibe was equal to ≈86,700 m³.

2.1.2 Description of the Project Activity

The project aims to generate renewable electricity through a 25MWp photovoltaic power plant located in Caraculo, Nambibe province, in Angola Southern Region, which currently relies mainly on diesel thermal plants for electricity production. The project is implemented by Solenova, a company jointly owned by Sonangol and Eni.

This project implementation represents the first phase of a larger photovoltaic project having a total capacity of up to 50 MWp. The overall project is implemented in following phases as requested by the Rede Nacional de Transporte ("RNT") as the sole offtaker, taking into consideration the current load and the need to ensure the full dispatchability of the energy produced. The first phase of 25 MWp will be followed by a second or more phases, up to 25 MWp as soon as the electricity load of the grid increases and RNT requires the additional photovoltaic plant ("PV plant") capacity.

The project activities that will address the sustainable development objectives, as mentioned in previous section, are, in particular, these three aspects:

- Supply clean electricity: the implementation of a photovoltaic system will supply clean energy to regional electricity network and will reduce the dependence on fossil fuel-based electricity, with benefits for air quality and reduction of GHG emissions;
- Technology transfer: the project aims to recruit local workers from near communities for the operational phase and maintenance of the plant, providing also training to workers and increase the awareness and knowledge among the local communities.

2.1.3 Description of the Project Activity in local language

O projecto visa gerar electricidade renovável através de uma central fotovoltaica de 25MWp localizada em Caraculo, província de Nambibe, na Região Sul de Angola, que actualmente depende principalmente de centrais térmicas a diesel para a produção de electricidade. O projecto é implementado pela Solenova, empresa detida conjuntamente pela Sonangol e pela Eni.

A implementação deste projeto representa a primeira fase de um projeto fotovoltaico maior com capacidade total de até 50 MWp. O projecto global é implementado nas seguintes fases, conforme solicitado pela Rede Nacional de Transporte (“RNT”) como única adquirente, tendo em consideração a carga actual e a necessidade de garantir a total despachabilidade da energia produzida. À primeira fase de 25 MWp seguir-se-á uma segunda ou mais fases, até 25 MWp, assim que a carga eléctrica da rede aumentar e a RNT necessitar de capacidade adicional da central fotovoltaica (“planta fotovoltaica”).

As atividades do projeto que abordarão os objetivos de desenvolvimento sustentável, conforme mencionado na seção anterior, são, em particular, estes três aspectos:

- Fornecer eletricidade limpa: a implementação de um sistema fotovoltaico fornecerá energia limpa à rede eléctrica regional e reduzirá a dependência da eletricidade baseada em combustíveis fósseis, com benefícios para a qualidade do ar e redução das emissões de GEE;
- Transferência de tecnologia: o projecto visa recrutar trabalhadores locais de comunidades próximas para a fase operacional e manutenção da fábrica, proporcionando também formação aos trabalhadores e aumentando a sensibilização e conhecimento entre as comunidades locais.

2.1.4 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
07/12/2020	Shareholder Agreement between Eni Energy Solutions, Sonangol and Solenova
30/10/2021	Final investment decision
20/10/2021	Environment impact assessment approval
02/11/2021	EPC Contract Award
30/03/2022	Concession agreement effective
30/03/2022	PV Power Plant construction
14/10/2022	Power purchase agreement signature
20/12/2022	PV Plant and substation building activities
15/12/2022	Site mobilization and preparation
30/01/2023	Environmental license for operation of the PV Plant
14/03/2023	Agreement of Substation Transfer between ENDE, Sonangol and Solenova
22/03/2023	Start date of the project activity, Commercial Operation Date (COD)
14/03/2023	Grid connection signature

2.1.5 Project Proponent

Organization Name	Solenova Ltd
Role in the Project	Implementator
Contact Person	Jaime Luzolo
Title	Managing Director
Address	Av. 4 de Fevereiro, 197, Torres Altântico, Luanda, Angola
Telephone	+244 222 637 440
Email	info@azule-energy.com

2.1.6 Other Entities Involved in the Project

Organization Name	Eni S.p.A.
Role in the Project	Authorized representative
Contact Person	Francesco Schio Eleonora Martinelli Nicola Fiore
Title	Carbon credits Portfolio Management Specialist
Address	20097 Via Emilia 1, San Donato Milanese (MI), Italy
Telephone	+39 0252042086
Email	francesco.schio@eni.com eleonora.martinelli@eni.com nicola.fiore@eni.com

2.1.6 Project Type

The project is a renewable energy project and is eligible under the sectoral scope “01 – Energy industries”, according to the VCS program. The project is not classified as a grouped project.

2.1.7 Project Location

The Project Area identified is located in Caraculo (Namibe Province), within the municipality of Bibala and adjacent to the “Estrada Nacional” EN 280 road. The selected area for the project is indicated in the polygon shown in the figure below and has an extension of about 165 ha, extending from the km 56+700 up to km 58+530 of the EN 280 road. The area is about 500 m a.s.l., with temperatures averaging between Tmax 42 °C and Tmin 9 °C, Tm 24,4 °C (Solargis).

The site is around 65 km from the capital of the Namibe province (Moçâmedes) and 117km from the capital of Huila province (Lubango). An existing 60kV power line, extended from Namibe to Lubango runs adjacent to the provincial road and to the selected area from east to west.

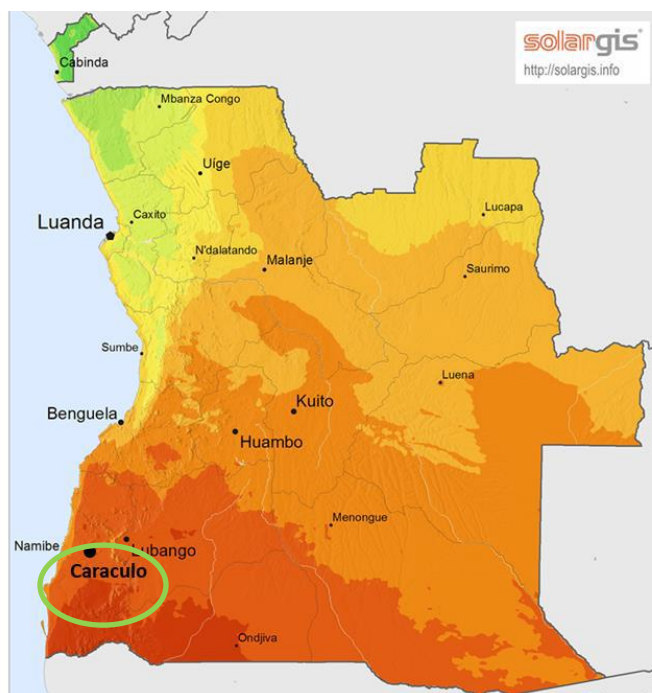


Figure 1: Project location



Figure 2 -3: Map of the PV Plant location

The coordinates of the above-mentioned identified area are reported below:

Table 2: Reference Coordinates of the project area

Coordinates	Latitude [UTM]	Longitude [UTM]
1	243960	8337950
2	245719	8338134
3	245983	8337898

4	245529	8337278
5	245178	8337028
6	243997	8337147

2.1.8 Baseline Scenario

The site is located in the Namibe Desert, and it is about two kilometers to the west of the Caraculo village. The area does not show any sign of residential or agricultural use, while it is currently partially used for grazing. The land is covered by spontaneous vegetation with small isolated trees and bushes. The project's prospect site is essentially flat, with some scattered outcrops.

The PV field will be connected to the Angolan Southern grid through a 60/15 kV substation to be built adjacent to the PV plant. As shown in the pictures below, the existing 60kV power line that goes from Namibe to Lubango runs adjacent to the provincial road and to the selected area from east to west. Linear distance between the substation and the existing 60 kV power line is about 200 m.



Figure 1: Project site area

The project will help in increasing the share of renewable energy in the Angolan energy mix. In absence of project activity, the site will continue current practice i.e. equivalent amount of electricity to be generated by the project activity would have been generated from other more fossil fuel intensive grid-connected power plants.

2.1.9 Causal Chain(s)

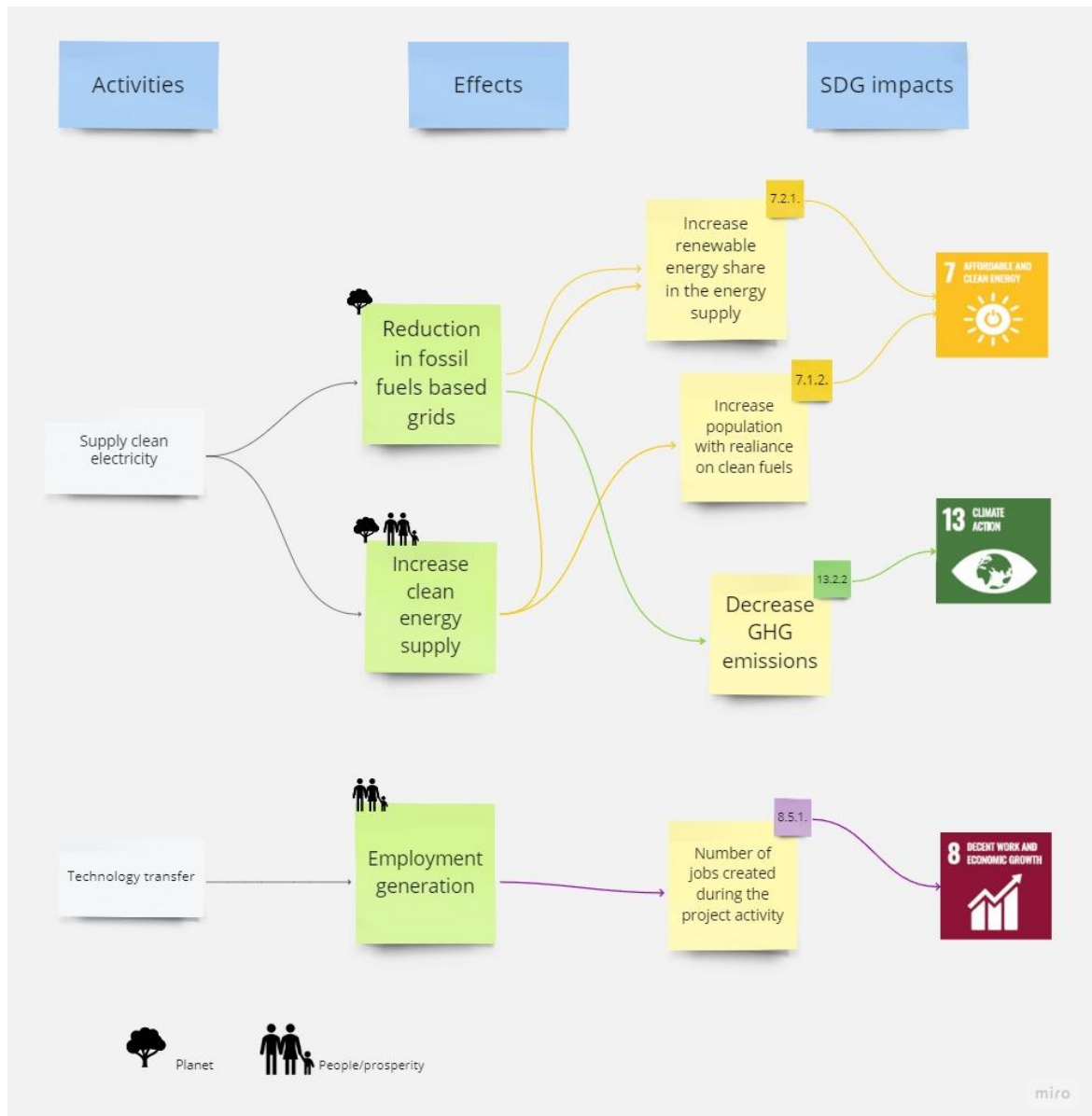


Figure 5: Causal Chain

2.1.10 Threats to the Project

- **Maintenance of Plant:** One of the risks of the project, which could slow down the implementation plan, is due to a lack of effective implementation of the plant or damage to its components due to external factors (such as extreme weather conditions).

Solution: In order to minimize any risk related to plant malfunction, an Operation and Monitoring plan (O&M Plan) is developed. Different elements can be identified in the Plan: (i) preventive maintenance; and (ii) corrective maintenance services. The first one has as its

main objective to ensure the reliability and stable operation over its lifetime, while the last one has the function of troubleshooting problems that may occur.

The O&M monitor the status of the Photovoltaic Plant helps to analyse incidents and energy production so that incidents that require corrective maintenance are rectified as soon as identified, and that maintenance is carried out in accordance with manufacturer's instructions and conditions. Additionally, O&M includes taking appropriate measures to minimize downtime derived from operational disturbances, in order to optimize the performance and maintain the Management of the Photovoltaic Plant informed about technical issues generated by the operation of the Plant, in order to extend its useful life.

- **Local acceptance:** any conflict with the local community can slow down or stop the project.

Solution: The identification, analysis and listening to stakeholders are fundamental elements for Eni: knowledge of expectations and transparency of relationships are the basis of the commitment to build trust and lasting relationships with its own stakeholders in Eni's areas of presence.

Eni has implemented a specific web-based platform called Stakeholder Management System (SMS) in order to carry out this daily and proactive engagement with stakeholders at local, national and international level. This tool allows to:

- map, classify and prioritize stakeholder and their requests pertaining to sustainability issues;
- monitor stakeholders involved in each project/area of presence, storing each interaction hold with them and analyzing their requests;
- highlight any critical issues and areas of improvement for the stakeholder engagement strategy to be developed (i.e. potential impacts on human rights, possible presence of vulnerable groups).

2.1.11 Benefit Permanence

Thanks to the Operation and Monitoring Plan described above, the project is supervised during its duration. The project proponent will supervise the financial requirements for project operation and will impart regular training to its employees to maintain benefits of the project. Once the duration of each solar plan is over, the proponent will replace them with new solar panels.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Identification

The identification, analysis and listening to stakeholders are fundamental elements for Eni: knowledge of expectations and transparency of relationships are the basis of the commitment to build trust and lasting relationships with its own stakeholders in Eni's areas of presence.

Eni has implemented a specific web-based platform called Stakeholder Management System (SMS) in order to carry out this daily and proactive engagement with stakeholders at local, national and international level. This tool allows to:

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- highlight any critical issues and areas of improvement for the stakeholder engagement strategy to be developed (i.e. potential impacts on human rights, possible presence of vulnerable groups).

Thanks to the information periodically uploaded and updated in SMS by Eni's business units which are using the tool, the SMS can be useful as a valuable source also for business projects in evaluation and selection phases located inside or near Eni's territories already available in SMS; in these cases, the most part of information on stakeholders is already present in the SMS. These data, suitably verified and adapted to the mentioned project, are completed by:

- analysis of desk studies and publications, including those focused on similar projects in the same country/region;
- analysis of any interactions with stakeholders which already took place in the previous months about the project;
- comparison and sharing of information with Eni's contact persons present in the area of the project area;

In this way, Eni implements a specific Stakeholder Register (SR), which includes:

- the list of stakeholders who are or could be affected / impacted by the project - both positively and negatively;
- a geographical map which shows the position of listed stakeholders

- the Relevance-Disposition Matrix, which prioritizes the stakeholders considered on the basis of their respective Relevance and Disposition towards the project;
- an analysis of any potential criticality that stakeholders could raise.

2.2.2 Stakeholder Description

Stk group	Relevance
National/Provincial/Regional Government	<i>High relevance given that the Project implementation plays an important contribute in achieving the energy transition target of the country.</i>
Host Communities	<i>Medium relevance given that the project takes place in an area with very low rate of urbanization, in an almost desert area far from the main villages.</i>
Universities/research centers	<i>Low relevance. Solar PV System is widely considered a positive technology for scientific communities.</i>
Business partners	<i>High relevance. The project is implemented through Solenova (a joint venture company between Eni and Sonangol).</i>

2.2.3 Stakeholder Consultation

Local institutional entities have been firstly contacted in April 2021 during the phase of preparation of the Environmental Impact Study of the plant and the substation. All relevant institution and authorities were contacted through dedicated correspondence to inform about the project and obtain relevant comments and information about project site and impacts. The following entities were contacted:

- Provincial Government Office,
- Provincial Office of Education,
- Provincial Office for Integrated Economic Development,
- Provincial Office of Infrastructure and Technical Services;
- Provincial Health Office,
- Provincial Office for Social Action, Family and Promotion of Women,
- Provincial Office of Agriculture and Fisheries,
- Provincial Office for Culture, Tourism and Environment,
- Protection and Fire Services.

In the annex section (Appendix 1) the correspondence that took place with the mentioned entities is reported. Few entities answered providing useful comments and information for the development of Environmental Impact Study and project development. All relevant local authorities were kept informed about activities throughout whole project development phase.

Regarding the local communities, a survey was performed in May 2021 with the purpose of informing the local communities about project activities and understanding their possible concerns as well as collecting various information about the necessary elements for understanding the current and future state of the locality, in the aspects, economic, territorial, cultural, etc., as per relevant legal requirement.

The survey investigates different aspect of the local community inhabitants: demographic information; housing condition and investigation about energy sources, water, electricity access etc; source of income and investigation about local economy and production; information about proximity services and their utilization (school attendance, hospitals or health centre); infrastructure and public transportation; local power relations and structure.

In the Annex section (Appendix 2) the questionnaires carried out are reported and, in the photos below, the moments of two interviews of the eleven interviews carried out are represented.

No relevant negative comments or critical recommendation about project implementation have been made by local stakeholders, which proved themselves in favor of the development of project activities.

For what concern the data gathering, information gathered at the local level (villages and neighborhoods) were integrated with information with the Deputy Communal Administrator and its Secretary of the Office, as well as contact with the Soba Grande of the town of Caraculo and its Secretary. Thus, the characterization was based essentially on the results of direct observation, and on the elements compiled at local level through informal conversations with the Communal Administration, Traditional Authorities and with the population.



Fig.6 Interview (Mbuanene province)



Fig.7 Interview (Duas Familias Province)

2.2.4 Continued Consultation and Adaptive Management

The plan chosen for the stakeholder engagement follows the precautionary principle of ensuring local approval by identifying and resolving the issues of concern to the local community as soon as possible.

The following methods for continuous consultation and grievance expression will be taken into account during the duration of the projects:

- Grievance expression process book
- Telephone access
- Internet/mail access

2.2.5 Anti-Discrimination

Eni fosters an environment of mutual respect where employees and business partners can work together without fear of harassment, discrimination, or infringement on human rights. Eni has strict policies against discrimination and in support of diversity and inclusion and any program implemented by Eni and its implementing partners ensures that all applicable laws, regulations, and international standards, including the Universal Declaration of Human Rights and core ILO Conventions, are adhered to. The United Nations Guiding Principles on Business and Human Rights (UNGPs) and International Conventions are key references for Eni for guiding activities, regulatory system and governance towards the respect for and promotion of human rights. Eni's commitment is reflected in Eni's Declaration on Respect for Human Rights and in a series of documents relating to Eni internal organisation and relations with external parties, such as the Code of Ethics, the Sustainability Policy and the Supplier Code of Conduct. Eni's approach to human rights is also confirmed in the international Global Framework Agreement on Industrial Relations and Corporate Social Responsibility, which was renewed in June 2019, and has an explicit reference to the UNGPs. Eni is also committed to the Voluntary Principles on Security and Human Rights (VPSHRs) and it was accepted as an "Engaged Corporate Participant" in the Voluntary Principles Initiative (VPI), the multi-stakeholder initiative involving governments, corporations and NGOs. In 2021, Eni also prepared an Annual Report describing the activities carried out during the year to implement the VPSHRs, a set of principles to support companies regarding respect for human rights in the management of security operations, defined and promoted by the VPI. Finally, Eni ranked first in the Corporate Human Rights Benchmark (CHRB) Human Rights Index in the latest study carried out in 2020, equally with just one other company.

2.2.6 Worker Training

Through the O&M Plan there are included frequent trainings to all employees regarding operational and management process. These training sessions increase the technological capacity and knowledge enhancement of the workers, building useful skills and keeping workers aware about safety issue.

The community members employed by the company receives continuous learning and training sessions through an ongoing process on maintenance and safety of handling equipment. These trainings help the employees recruited from communities to build new skills and develop their capacity and knowledge.

2.2.7 Equal Work Opportunities

Eni approach has always been rooted in the international culture of plurality. It is based on the fundamental principles of non-discrimination, equal opportunity and inclusion in respect of all forms of diversity, as well as integrating and balancing work with personal and family needs. Eni has chosen to discuss these issues with both national and international peers by joining organisations and participating in inter-company initiatives and benchmarking systems for sharing best practices and identifying common activities in the area of Diversity & Inclusion. They are the European Round Table (ERT), World Economic Forum (WEF), Industrial ALL Global Unions and the UN Global Compact, among others. Eni has also signed up to the Women Empowerment Principles, the gender equality and women's empowerment initiative of UN Women and the UN Global Compact.

2.2.8 Workers' Rights

Eni is committed to respect and promote internationally recognized workers' rights in all the Countries where it operates. This commitment goes beyond the International Bill of Human Rights and the International Labour Organization (ILO) Declaration on Fundamental Principles and Rights at Work, which together constitute the minimum requirements of businesses according to the UN Guiding Principles, and includes also:

- the development of a comprehensive framework of internal policies and rules to affirm the commitment to respecting internationally recognized workers' rights that could be most relevant to Eni's business.
- The commitment of Business Partners and suppliers to also comply with them.
- The collaboration with unions and the engagement with international organizations including the ILO.

In April 2020, Eni published the Supplier Code of Conduct: it establishes the mutual commitment to recognize and protect the value of all the people, commitment to contrasting climate change and their effects, operating with integrity, protecting company resources, promoting the adoption of these principles within their own people and supply chain. Among these principles, regarding human rights and work, the reference is the International Labor Organization (ILO) Declaration on Fundamental Principles and Rights at Work. All suppliers will be asked to sign the Code and to promote the principles contained in it along their supply chain. Specific commitments on human rights are:

- Prohibit the forced labor, the undeclared labor, the compulsory labor and all the forms of modern slavery and human trafficking.

- Prevent any form of work by children under the age of 15 and ensure, in compliance with local law, that teenagers under the age of 18 are not employed in hazardous jobs.
- Abide by working times and rest periods in compliance with the applicable legislation and in line with international standards.
- Prevent any kind of discrimination.
- Ensure respect of workers' rights and trade unions freedoms.
- Establish clear and fair working conditions defined in the employment contract.
- Respect the cultural, economic and social rights of the local communities, minorities, indigenous peoples and other vulnerable groups.

In line with Eni corporate culture, which promotes inclusion and respect for uniqueness, in December 2021 Eni issued a policy against violence and harassment in the workplace. The Zero Tolerance policy, in accordance with the principles of the Geneva Convention of the ILO, defines a broad perimeter of types of harassment that allows to identify misconduct and behaviour that should not be engaged in and should be reported.

2.2.9 Occupational Safety Assessment

In order to apply the general “Health, Safety and Environment” criteria and minimum requirements to the photovoltaic project design, complying also with local laws and regulations as well as with international codes and standards, a specific document called “HSE Philosophy” has been prepared for this project.

The basic objective of this philosophy is to ensure an inherently safe design of the plant in order to:

- minimize the potential of hazardous events;
- minimize the possibility of accidents/injuries to personnel and reduce the potential for loss of life;
- minimize, during the operation phase, the risks to workers' health;
- minimize the risk and consequences of an accidental event;
- ensure a safe working environment for personnel;
- ensure adequate means of escape;

- provide sufficient safety devices;
- provide appropriate fire protection systems to rapidly bring under control and extinguish any reasonably foreseeable fire which could develop during normal operations;
- identify and assess any potential impact on the surrounding environment and to ensure that these are avoided and/or minimised.

In the document, specific sections are dedicated to workplace layout (offices, worker dormitories, canteen) , to the handling requirements of loads, materials, equipment, to the protection against direct and indirect contacts, fire protection system and other risks.

2.2.10 Feedback and Grievance Redress Procedure

The feedback and grievance procedure ensures that every issue is properly prioritized and addressed. In the figure 8 below there is a schematic description of the management of grievances.

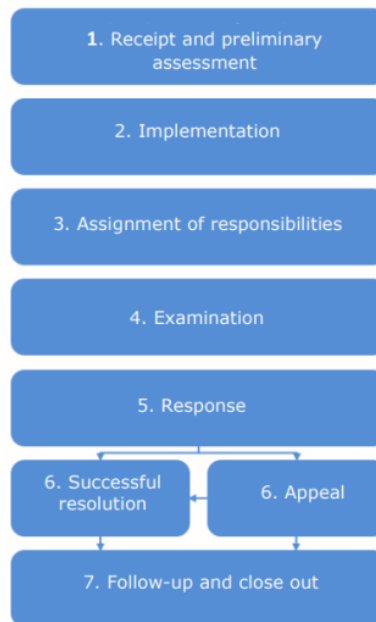


Fig 8 Management of grievance process

As a first step, the procedure involves a preliminary assessment, that includes a preliminary classification and prioritization of the grievance. The person (or group of persons) who expressed a grievance is promptly notify that it has taken charge of it. Together with the acknowledgement of receipt of the compliant, it is indicated also, where possible and appropriate, the activities

planned for the management of grievance and the timeframe that can reasonably be expected for the conclusion of the process. The preliminary assessment also involves the assignment of responsibilities and of verification to the proper functional units in order to examine the grievance. In parallel with the ongoing examination procedure, the complainant is kept adequately informed about the progress of the procedure. Once a proposed resolution has been approved, the response is notified and discussed with the complainant, that may agree (successful resolution) or disagree. In the latter case, the complainant response is examined again and it may be possible the involvement of a third party to resolve the grievance.

2.2.11 Feedback and Grievance Redress Procedure Accessibility

During the consultation, the stakeholders had the possibility to ask clarifications and expose grievances/opinions/suggestions. A dedicated phone number and email address was shared in case of further information needed or in case of an issue to be solved and are always operated through the lifetime of the process. Moreover, also through the engagement and monitoring process, the beneficiaries are asking to provide feedback and there is the possibility to assess grievances.

2.2.12 Stakeholder Access to Project Documentation

During stakeholder consultations, all the project details were shared among participants. Considering that many project documents are very technical, a comprehensible non-technical summary detailing the project activities and outcomes were shared during the public consultation, together with the phone and email contact to reach field staff implementors for further clarification. All the project details are available either at the centralized level and at the local office in the host country.

2.2.13 Information to Stakeholders on Assessment Process

Stakeholders will be informed of SD Vista process, including validation and site visit: prior notice will be always granted giving the possibility to interested people to be available.

2.3 Project Management

2.3.1 Avoidance of Corruption

Eni considers corruption to be an intolerable obstacle to business efficiency and fair competition. In November 2009, to implement this commitment in the business, the company voluntarily extended its compliance program to the risks of corruption by adopting the Anti-Corruption Guidelines and their Ancillary Procedures. Furthermore, Eni's Whistleblowing reporting channel allows employees, members of corporate bodies or third parties to forward (including confidentially and/or anonymously) complaints relating to problems in the internal control system (compliance with law and corporate policies, guidelines or procedures etc.) or other violations of the Code of Ethics (issues involving ethical conduct, mobbing, harassment, conflicts relating to human resource management, etc.). The Anti-Corruption Compliance Program is rooted in the existing national and supranational regulations concerning corruption and money laundering and, within Eni, is embodied, from a regulatory point of view, in the "Anti-Corruption Management System Guideline" as well as in other detailed regulatory instruments (so-called Anti-Corruption Regulatory Instruments) which constitute the reference framework for identifying activities at risk and the control instruments that Eni makes available to its People to prevent and combat the risk of corruption and money laundering. The Compliance Program is mandatory for Eni and all its Subsidiaries, in Italy and abroad. Eni also recognises the importance of clearly communicating its values and ethical principles to its counterparties/partners, providing for adequate anti-corruption and anti-money laundering compliance commitments in the relevant contracts. Eni also considers transparency a corporate value and views its promotion as crucial for a more inclusive resource governance in favour of communities' interest and as a prerequisite to fight

international corruption. Eni's commitment is confirmed by active participation, since 2005, to the voluntary multistakeholder initiative of EITI (Extractive Industries Transparency Initiative) and by the support to its efforts in promoting an open and accountable management of natural resources.

2.3.2 Statutory and Customary Rights

The project has all the legal, customary rights on the land and does not require any change to land tenure arrangements because the project activities is implemented on private lands. Therefore, the project proponent did not require any clearances from nodal agencies.

2.3.3 Recognition of Property Rights

This section is not applicable since the project does not require any change in property rights.

2.3.4 Free, Prior and Informed Consent

This section is not applicable since the project does not require any change in property rights.

2.3.5 Restitution and/or Compensation for Affected Resources

This section is not applicable since the project does not require any change in property rights.

2.3.6 Property Rights Removal/Relocation of Property Rights Holders

This section is not applicable since the project does not require any change in property rights.

2.3.7 Identification of Illegal Activities

There are no illegal activities identified and associated that could affect the project's impacts and hence measures needed and designed to reduce these activities are not required.

2.3.8 Ongoing Conflicts or Disputes

This issue is not applicable since the project activities do not affect property rights or property disputes.

2.3.9 National and Local Laws and Regulations

In order to differentiate its economy and to meet the energy demand of its growing population, Angola government launched a long-term plan for the energy sector, called "Angola Energy 2025". The Angola 2025 strategy's main goal is to provide access to basic energy services to the population.

Eni, in cooperation with Angolan O&G Company Sonangol, are pursuing the opportunity to develop projects to exploit the Angolan renewable potential, in line with their strategy to expand their activities to zero carbon initiatives and support Angola in the diversification of its energy supply matrix.

In this context, a Memorandum of Understanding has been signed on November 27th, 2017, between Eni and Sonangol setting forth the general terms of parties' cooperation for the development of several initiatives, including renewable energy projects. In 2019, Eni and Sonangol signed in Luanda the Heads of Agreement for the incorporation of Solenova Ltd, a joint controlled company set up to assess and develop renewable energy opportunities in Angola.

The Carcaculo photovoltaic plant is the first business opportunity jointly identified in the renewable energy sector. The project activity meets all relevant national and local laws and regulations, as listed below.

For Civil, Structural and Architectural Works:

- EN Eurocodes 0, 1, 3 and 9 for Structural Analysis and Design;
- ISO 12944 for Corrosion Protection of Steel Structures.

For Electrical, Instrumentation and Telecommunication:

- IEC 60296, IEC 60156, IEC 60422, IEC 60475 for the Transformer;
- IEC61850, IEC 60870 for the PCCS of the Substation;
- IEC 62271, IEC 60529, IEC 60470, IEC 60071, IEC 60282, IEC 61243, IEC 61869, IEC 60721 for the switchgear;
- IEC 62271-203, IEC 62271-100, IEC 62271-102, IEC 61869 for the GIS;
- IEC 60076-1 for the auxiliary transformer.

The project successfully obtained the following main permits and authorization from national and regional authorities, demonstrating compliance with relevant legal requirements:

- Land Concession Title;
- Study of Technical Reference Conditions (ECTR);
- Project Execution Plan;
- Environmental Impact Assessment (EIA);

- Environmental License for Installation;
- Environmental License for Operations;
- Establishment License;
- Registration of the Project to AIPEX (Agency for Private Investment and Promotion of Exports of Angola);
- Comercial Alvara;
- Registration to INE (Instituto Nacional de Estatística);
- Concession Contract for Construction and Exploration of the PV Plant.

2.3.10 Project Ownership

Solenova Ltd is the owner of the project by means of Shareholders' Agreement and has the legal right to control and perform the project activities.

2.3.11 Grouped Projects

This section is not applicable.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Condition of Stakeholders at Project Start

According to the Environmental and Socio Impact Analysis of the project, these are the main socio economic and cultural features of the stakeholders living in Namibe Province:

- **Administration:** The Namibe Province is composed of 5 municipalities, 15 communes, 44 villages in urban areas and 353 villages in rural areas.
- **Demography:** The age structure of the population shows that the Province of Namibe is made up mostly of by an extremely young population, the average age of the population being around 20 years, both for men and women. The proportion of the population aged 0-14 is 48.2%, and that of the elderly (65 years or older) is only 2.1%, values of the same order of magnitude than the national average.
- **Employment:** The labor market concentrates around 44% of the population aged 15 or over, with activities in the primary sector (Agriculture and Fishing) account for 32%, and those in the secondary sector (Industry, Construction and Energy) account for 7%, while activities in the tertiary sector (Commerce and Services) account for 37%. However, 24% of the resident population did not declare the its economic activity. The unemployment rate was high in 2014, covering 18.0% of the active population, with greater incidence in the urban population (92.2%) and much less in the rural population (20.3%). These values are globally lower than the national average (24.2%), but unemployment in the urban population is much higher than that observed in the country, which was 30.8%)
- **Education:** The rate of literacy (expressed as literate people with respect to the total population older than 15 years), in Namibe Province is 64,3% distributed as: 88,3% for urban population and only 29.1% in the rural population. For the age range 15-24 years, it is possible to observe a significant improvement, with 75.2% of the population that know how to read and write (91.4% of urban population and 38.3% of rural population).
- **Services:** access to suitable drinking water covers around half of total households in the province; only about 41% of households have appropriate sanitary facilities; access to electricity from the public network covers about half of total population in the province, with a large incidence in the urban area (75%) and only 4% in the rural households.

- Ethnic group: The overwhelming majority of the population considers itself to belong to the ethnolinguistic group of “Herero”, a group including a few thousand people scattered between Namibia, Botswana and a part of Angola.

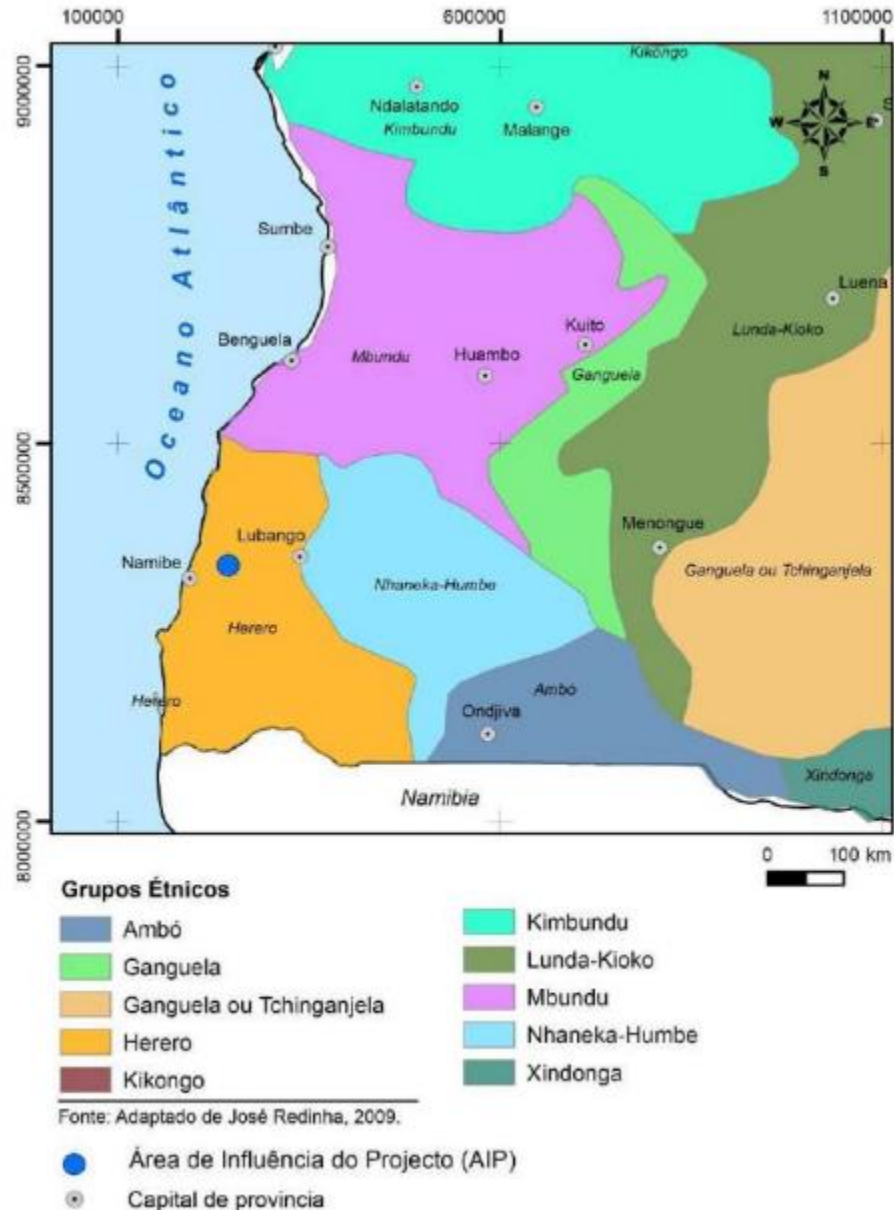


Fig. 8 Ethnic group of the region

In the project area, two primary stakeholders groups have been identified, with homogeneous baseline scenarios and project outcomes:

1. Local community members: people living in the villages of the area, involved in agricultural sector and lacked technical expertise in electrical domain; they rely on non-renewable energy sources;
2. Small business/shop owners: Due to the limited availability of grid electricity, small businesses and shop owners had to curtail their operations in the evening, resulting in reduced income and business activities.

3.2 Expected Impacts on Stakeholders

Impact #1	Local Economy
Type of Impact	Positive, actual, direct and indirect
Affected Stakeholder Group(s)	Raw materials Producers and other services in the town of Caraculo
Resulting Change in Well-being	With regard to boosting the local economy, the activities necessary for the functioning of the Photovoltaic Plant and its maintenance involve the acquisition of different materials (such as raw materials, lubricants, among others) and services, including maintenance of access. These activities will benefit the local economy, especially the town of Caraculo, and may attract other investments, with positive effects on the population and economic activities, being a positive, certain, direct and indirect, local, permanent, immediate impact even if small magnitude and moderate importance.

Impact #2	Increase availability of electricity
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Shop owners and small business

Resulting Change in Well-being	Shop owners can be able to operate their shops as per their requirements, without the constraints of low light conditions such as early morning and late evenings, as well as during nighttime. This enable them to maximize their business hours and increase their revenue. Small businesses can continue their work uninterrupted, without any disruptions caused by power outages. This creates a stable and reliable source of electricity, which is crucial for businesses to function efficiently and sustainably.
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Impact #3	Local Employment
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local workers
Resulting Change in Well-being	The construction of the project will generate a local demand for labor in the civil construction sector, contributing to temporarily mitigate the unemployment levels. Moreover, during the operation phase of the Photovoltaic Plant it will be necessary the existence of a technical team for assistance in terms of surveillance and maintenance and, for that, permanent jobs are created.

Impact #4	Enhance skills and employment
Type of Impact	direct and positive
Affected Stakeholder Group(s)	Residents employed by the project activity
Resulting Change in Well-being	The project activity has created many employment opportunities for people from the surrounding villages as operations and maintenance staff. These individuals have received comprehensive training and capacity building workshops to equip them with the necessary skills to operate and maintain renewable energy power plants. This is particularly significant, given the high levels of unemployment in the area and the fact that most residents are employed in the agricultural sector.

3.3 Stakeholder Monitoring Plan

The monitoring Plan represents a key element to improve the current and future management of the outcomes of the stakeholder engagement process. The Project Proponents systematically monitors and evaluates the level of project's objectives for stakeholders.

The data and information collected about the project are used to guarantee:

- The project progress state check
- The planning of any corrective measures to be applied during the implementation phase
- The elements for evaluating the effectiveness of implemented measures and the long-term impacts derived from them for the stakeholders.

The analysis and classification of the collected information from stakeholder may lead to a request for corrective measures or additional data

The stakeholder impacts will be evaluated through the assess of primary survey associated with the implementation of project activities.

Stk group	SDG INDICATOR	Monitoring Parameter	Parameter description	Source
Beneficiaries	7.1.2 Increase population with realiance on clean fuels	Number of people/households served by the project	Number of people/households served by the project > baseline condition	Plant records for energy generation
	7.2.1. Renewable energy share in the toal final energy consumption			

Beneficiaries	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities.	Number of local people involved in the project	Local job occupation > baseline condition	HR department records
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3.4 Net Positive Stakeholder Well-being Impacts

All the impacts described in section 3.2 are positive, therefore the net impacts of the project activities are positive.

In particular, for each stakeholder group:

Stk group	Impacts	Baseline scenario	Project scenario	Net impact
Small business	#1 Local Economy growth #2 Increase availability of electricity	Initially, due to the limited availability of grid electricity, small businesses and shop owners had to curtail their operations in the evening, resulting in reduced income and business activities.	Under project scenario, the supply of electricity will increase and the investment to access to electricity.	Positive

Local communities		People living in the villages of the area, involved in agricultural sector and lacked technical expertise in electrical domain; they rely on non-renewable energy sources	During the construction phase and during the operational phase, the project will generate demand of local workers and enhance their technical skills with activity training.	Positive
	#3 Local employment increasing #4 Enhance skill and employment			

4 BENEFITS FOR THE PLANET

4.1 Condition of Natural Capital and Ecosystem Services at Project Start

The characterization of the territory feasible for the Project was carried out based on the interpretation of satellite images (Google Earth software), measured by collecting elements during the field visit carried out in April and May 2021. The baseline information for the characterization of ecosystem services was obtained through available secondary information (bibliography) for the Province of Namibe and primary information obtained from the local population by the project component team socioeconomics.

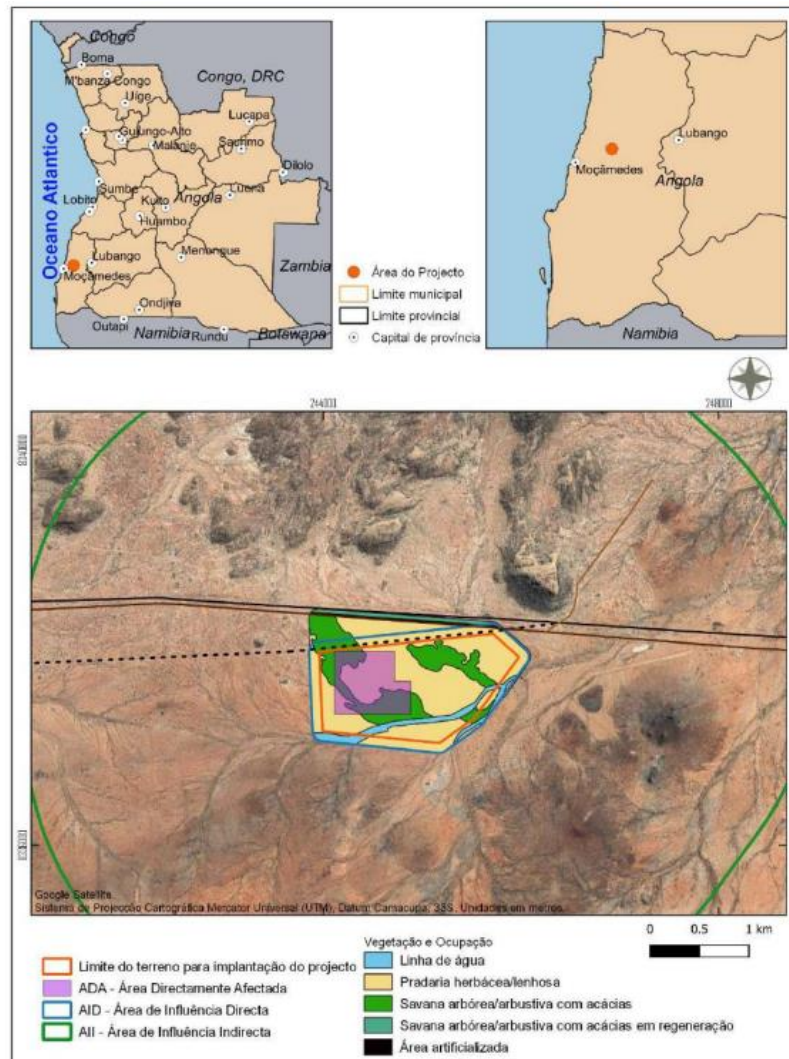
- Morphology: The interested area is inserted in a vast area of relief relatively plan. This coastal strip presents altimetric quotas of the order of 500 m. All existing water courses in the vicinity are intermittent, depending heavily on rainwater that occurs very sporadically, which translates into the climate of the region as arid and desert.



Fig.8

- Soils: In the area allocated to the Project there are arid soils and scattered rocky outcrops in the surroundings: although it does not allow the proper specification, they are poorly evolved soils that relate to the sandy covering deposits. They are made up of materials with a coarse texture and a low percentage of clay, constituting themselves as loose soils and very little differentiated. Thus, they are generally not very fertile, with little aptitude for agricultural and used mainly as pasture, since they promote the rapid infiltration of water and a reduced capacity to retain nutrients and organic matter. The land is covered by spontaneous vegetation with small, isolated trees and bushes. The presence of bare soil is frequent in all plant formations, due

to the occurrence of grazing. As it is possible to observe in the following figure, there is only an area artificialized (black line, corresponding to the route EN 280).



- **Climate:** The climate of the Namibe Province is characterized as arid along a wide western and semi-arid in the remaining part. The Project area falls within a territory with arid climate (potential annual evapotranspiration greater than annual precipitation), desert (average total annual precipitation of about 200 mm), dry and hot (average temperature annual temperature above 18°C), with a short wet season (about two months) and a prolonged dry season, with warm average temperatures.



- No threatened species have been confirmed in the interested area, according to the IUCN (International Union for the Conservation of Nature, 2021), nor species included in the Red List of Angola.
- Heritage: In the area of direct influence of the project, no natural or anthropic resource was identified that can be considered as cultural and/or natural heritage, monument or site of cultural value.

4.2 Expected Impacts on Natural Capital and Ecosystem Services

Impact #1	Soil sealing
Type of Impact	Negative, direct
Affected Natural Capital and/or Ecosystem Service(s)	Soil
Resulting Change in Condition	It is during the construction phase that most of the negative impacts on the soils, considering that it is in this phase that the constructive actions of the Project will take place, namely: i) clearing, stripping and excavation of topsoil; ii) movements of soils; iii) circulation of vehicles and machinery assigned to the work; iv) installation of the shipyard and use of the support structures for the work. considering that the nature of the interventions does not determine significant allocations (maximum excavation depth less than 10 m), where the relief present does not constitute areas at risk of erosion, it can be concluded that the impacts associated with mentioned actions will be negative, certain, direct, irreversible and permanent (in the areas that will be occupied by the Project's infrastructure) and reversible and temporary (in other areas), globally constituting impacts of reduced magnitude. It is not foreseeable that the soils suffer important impacts in the operation phase, since there is a need to intervene in any new areas, nor to circulate or carry out any other type of operations outside the accesses and assembly areas established during the construction phase. In the operation phase there will also be a reduction in the area affected in the construction phase, which corresponds to the recovery of the areas occupied by the shipyard and the areas ancillary to construction. As for the significance of the impacts, this will be related to the suitability of the affected soils, which will be the more significant, the greater their agricultural aptitude. In this sense and considering that the interventions are carried out in soils that are not suitable for agriculture (soils dry), it can be concluded that the negative impacts of deforestation and pickling operations of topsoil on present soils will be globally of low importance.

Impact #2	Reduction polluted emissions
Type of Impact	Positive, indirect
Affected Natural Capital and/or Ecosystem Service(s)	Air quality
Resulting Change in Condition	The photovoltaic project has a total capacity of up to 50 MWp to be developed with a phased approach: a first phase of 25 MWp and a second or more phases of 25 MWp, that represents a strong contribution to the installed power in Angola from renewable energies. In this context, the indirect positive impacts on air quality can be considered resulting from the production of energy through the use of a non-polluting renewable source, in to the detriment of the burning of fossil fuels, thus contributing to the minimization of equivalent CO2 emissions. The production of energy without recourse to greenhouse gas emissions greenhouse, will imply a positive impact, certain, indirect, permanent, immediate and scope regional, which gives it a medium magnitude. This impact is considered to have high importance within the scope of the global strategy to combat the issue of climate change/reduction of greenhouse gas emissions. In the operational phase of the Project, with the exception of occasional maintenance actions, it is not expected road traffic associated with the project, which is also positive in terms of emissions.

Impact #3	Waterproofing
Type of Impact	Negative, direct
Affected Natural Capital and/or Ecosystem Service(s)	Soils

Resulting Change in Condition	Although the area under the panels will be recovered after completion of works with regeneration of the vegetation, there will be a partial waterproofing of the soil in the area under the panels that will be protected from the direct impact of precipitation. However, the runoff of rainwater in the panels will concentrate the arrival of that water to the soil and from there it infiltrates the ground that meanwhile started the vegetation regeneration process. It is assumed, therefore, that in an initial phase the runoff takes place more quickly, making infiltration more difficult, but after a year or so two, the infiltration of water that runs on the surface of the panels is completely infiltrated, restoring the slow infiltration of water into the soil. It is therefore considered that the negative impact of the reduction temporary infiltration capacity is of reduced magnitude and low importance because it is not considered that it will affect the overall recharge of the aquifer system.
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Impact #4	Climate
Type of Impact	Positive
Affected Natural Capital and/or Ecosystem Service(s)	Soils and air

Resulting Change in Condition

During the operating phase, microclimatic changes may occur due to the insulating effect of the solar panels, which reduces heat loss from the ground surface, and their impact on the speed of wind and turbulence, altering the distribution of heat and water vapour. However, it is recognized by some authors that solar parks will have different impacts depending on their location. Less solar radiation under panels in regions of greater insolation, as is the case of Caraculo, it can even be positive and boost plant growth.

There is a set of potential direct microclimatic impacts and, despite the scarce knowledge on this matter, which prevents a precise quantification of impacts, it can be said that in the study area these impacts are expected to be of magnitude negligible.

Additionally, there are indirect positive impacts on the climate resulting from the energy production through the use of a non-polluting renewable source, to the detriment of the burning fossil fuels, thus contributing to the minimization of CO2 emissions equivalent. The production of energy without recourse to the emission of greenhouse gases, will imply a positive, certain, indirect, permanent, immediate and regional impact, which gives an average magnitude. This impact is considered to be of high importance in terms of scope of the global strategy to combat the problem of climate change.

4.3 Natural Capital and Ecosystem Services Monitoring Plan

Project Control Monitoring is a task that will be performed continuously during the Execution phase of the Project while Reporting will be executed on a predefined frequency and/or on specific request. The Project Control System will be able to monitor the execution of the project activities on their schedule and cost components and determine any possible variation of the agreed plans.

SDG INDICATOR	Monitoring Parameter	Parameter description	Source
13.2.2	Total greenhouse gas emissions per year	Reduction of GHG emissions for energy production and supply respect to a baseline condition	Plant records for energy generation

According to the VCS project ID 4026 to which this SD VISta project document is linked, the total greenhouse gas emissions per year are monitored and computed through the methodology “ACM0002: Grid-connected electricity generation from renewable sources”, v21.0.

As per the approved consolidated Methodology ACM0002 (Version 21) para 47: “Baseline emissions include only CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants”. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (t CO2/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO2 emission factor for grid connected power generation in year y calculated using TOOL07 (t CO2/MWh)

Monitoring objective

As per ACM0002 (Version 21), the emission reductions achieved by the project activity will be determined ex-post through direct measurement of the amount of net electricity supplied to the grid multiplied by the combined margin emission factor:

$$ER_y = EG_{facility,y} \times EF_{grid,CM,y}$$

Since the emission factor is fixed for the first crediting period, the aim of the monitoring is only to monitor the quantity of net electricity generation supplied by the project power plant to the grid using electrical energy meters. The project proponent uses electricity from the grid during the period when the PV plant is not producing solar electrical energy, so both electricity consumption and generation will be monitored:

$$EG_{facility,y} = EG_{Export,y} - EG_{Import,y}$$

Electricity meters

Three phase smart meter P2000 series are MID approved with Annex B and Annex D, KEMA certified per IEC62053-21/22/23 and EN 50470 standards, and DLMS certified as LN HDLC & TCP compliant.

Two models of the P2000 meter are available: transformer operated model P2000-T, and direct connected (whole-current) model P2000-D. Both models are available in accuracy class of EN 50470 Class A, B, and C, or IEC 62053 Class 1.0, 0.5S, and 0.2S.

Energy Metering

1. Energy measurement using digital measuring chips, therefore metering precision can be digitally verified.
2. High overload-multiplier energy metering (12 times multiplier metering for class 0.2S or 0.5S) by applying variable gain measurement and multi-section digital compensation.
3. Active energy, reactive and apparent energy metering through data provided by energy measuring integrated chips. 4-Tariffs (default, up-to 8 tariffs optionally) and total energy metering of active energy, reactive and apparent energy. Energy displays up to 11 digits, with decimal points configurable.
4. Energy displays up to 11 digits, decimal points configurable from 0 to 3 digits. User can select to display Primary values or Secondary values. Energy unit is also programmable as 'k' or 'M', i.e. kWh/MWh, kvarh/Mvarh, kVAh/MVAh.
5. Pulse indication of energy metering using two LED's, which indicate active energy kWh and reactive energy kvarh. The duty cycle of LED output is 50%. 4 Channel of impulses output, the pulse width is 80 milliseconds by default(optional).
6. kWh and Kvarh per Phase: P2000 measures per phase import and export active and reactive energy including:
 - ✧ Phase A +kWh, -kWh, +kvarh, -kvarh
 - ✧ Phase B +kWh, -kWh, +kvarh, -kvarh
 - ✧ Phase C +kWh, -kWh, +kvarh, -kvarh
7. Meter can keep on operating and recording as long as voltage exists on any two terminal of the meter with or without neutral.

All these values can be programmed to display on LCD or read from communication ports.

Instrumentation Data

1. P2000 measures instantaneous phase voltage, phase current, power factor (total, and per phase). The data can be read through optical port, or GSM/GPRS, or displayed directly on LCD.
2. P2000 measures instantaneous value of phase angles.
3. The active instantaneous power kW (total and per phase), reactive instantaneous power kvar (total and per phase), and apparent power kVA (total and per phase) per second can be read through optical port, or displayed on LCD.
4. Instantaneous value of Frequency (Hz).
5. P2000 measures instantaneous up to 31st order Current, Voltage Harmonic Components.(optional)

Technical Data

Specification

Connection:	3 Phase 4-Wire
Voltage:	3x230/400V, 3x63.5/110V
Current:	3x1(10) A, 3x10(160) A
Frequency:	50 Hz or 60Hz
Work temperature:	-25°C to 70°C

Voltage circuit consumption:

Voltage	Voltage burden (VA)	Voltage burden (W)
63.5V	<3.0	<1.5
100V	<3.0	<1.8
230V	<4.0	<1.8

Current circuit consumption: <1VA (I_b = 10A)

Maximum Demand Indication

Calculation method: Sliding or block mode

Calculation cycle: 5, 10, 15, 30, or 60 minutes

Rolling interval: 1, 2, 3, or 5 minutes

Recording range: 99.9999 kW

Additional error: $\pm (0.5+0.05 P_m/P_n) \%$

P_m - power value at I_{max} (cosφ=1.0)

P_n - equivalent real power value on demand indicator

Data Monitoring & Management Procedures

The monitoring is done according to the following procedures:

- Data from electricity meters are recorded at project site by project proponent;
- On monthly basis Project proponent provides to the Buyer a reconciliation act for agreement and signing together with invoice for payment on the actually supplied electricity.
- The Buyer confirms the electricity amounts on the receipts;
- Financial Settlement Center pays the money to project proponent for the confirmed export amount of electricity.

Quality assurance and quality control procedures

An internal Project Quality plan has been established for the project to define how internal requirements and activities shall be carried out.

The QA/QC plan shall be distributed and implemented within the project organization, and frequently updated based on progress and changes in the project. Where a requirement results in a deviation from the management system, the deviation shall be handled in accordance with internal non-conformance and deviation procedure. The quality plan describes the management processes to be carried out during the project lifetime. The QA/QC plan is one of the main governing documents for the project.

4.4 Net Positive Natural Capital and Ecosystem Services Impacts.

As a renewable source of power, solar energy has an important role in reducing greenhouse gas emissions and mitigating climate change, which is critical to protecting humans, wildlife, and ecosystems. Solar energy can also improve air quality and reduce water use from energy production.

The project first phase implementation (25 MWp) will permit a diesel consumption reduction of about 18.000 m³/year. The average annual GHG emission reduction expected with the implementation of the project activity is equal to 40.252 tCO₂e per year.

In line with the Angola 2025 strategy and international commitments, the priorities defined within the scope of the electricity policy are the following:

- Increase the average electrification rate at national level and reduce access asymmetries throughout the territory;
- Optimize the location and availability of existing thermal generation, as well as increase installed power, namely in new and renewable energies (Solar, Wind, Biomass and Mini-Hydric), aiming at replacing fossil fuels;
- Progressively replace public investment in electricity generation with long-term private funding, reserving public funding for structural investments.
- Enable the expansion of access to electricity to many municipal seats throughout the territory, through the establishment of partnerships between Municipal Administrations or the National Electricity Distribution Company (ENDE)

From this framework, it clearly appears that the project is in line with the national strategy defined for the sector and will contribute 25 MW out of a total of 800 MW to be installed by 2025 to the National goal.

Finally, reference is made to the indirect and likely positive impacts of its implementation:

- Improvement in security, reliability and supply of electrical energy – the production of energy from renewable sources is of considerable importance to supply the energy sector. Decentralization in the supply of electrical energy can configure less dependence on conventional sources, increasing reliability and, therefore, improving the local energy supply.

- Increase in equipment efficiency – the greater applicability of photovoltaic systems tends to increase investments in Research, Development and Innovation projects, which, consequently, would require greater levels of efficiency of the equipment components of these systems, in order to increase their viability technically and economically, thus contributing to the development of this technology.

With respect to the impacts described in the section above, here below a summary: since all the positive impacts are greater in magnitude respect to the negligible negative impacts, the net impact of the project on natural capital and ecosystem is positive.

<i>Impacts</i>	<i>Baseline scenario</i>	<i>Project scenario</i>	<i>Magnitude</i>	<i>Net impact</i>
#1 Soil Sealing	The area does not show any sign of residential or agricultural use, while it is currently partially used for grazing. The land is covered by spontaneous vegetation with small isolated trees and bushes. The project's prospect site is essentially flat, with some scattered outcrops.	Temporary negative impact during construction phase: (i) clearing, stripping and excavation of topsoil; ii) movements of land; iii) circulation of vehicles and machinery assigned to the work; iv) installation of the shipyard and use of the support structures for the work. It the operations phase of the project is not foreseeable that the soils suffer important impacts in the operation phase, since there is no need to intervene in any new areas, nor to circulate or carry out any other type of operations outside the accesses and assembly areas established during the construction phase.	Low	Negative

#2 Reduction polluting emissions	The area does not show any sign of residential or agricultural use, while it is currently partially used for grazing. The land is covered by spontaneous vegetation with small isolated trees and bushes. The project's prospect site is essentially flat, with some scattered outcrops.	The indirect positive impacts on air quality can be considered resulting from the production of energy through the use of a non-polluting renewable source, into the detriment of the burning of fossil fuels, thus contributing to the minimization of equivalent CO2 emissions.	High	Positive
#3 Waterproofing	The interventions are carried out in dry and arid soils that are not suitable for agriculture.	Partial waterproofing of the soil in the area under the panels that will be protected from the direct impact of precipitation. However, the runoff of rainwater in the panels will concentrate the arrival of that water to the soil and from there it infiltrates the ground that meanwhile started the vegetation regeneration process. It is assumed, therefore, that in an initial phase the runoff takes place more quickly, making infiltration more difficult, but after a year or two, the infiltration of water that runs on the surface of the panels is completely infiltrated, restoring the slow infiltration of water into the soil.	Low	Negative

#4 Climate	The Project area falls within a territory with arid climate desert, dry and hot	Negative impact on microclimatic changes: loss of surface, turbulence altering the distribution of heat and water. Magnitude negligible. Indirect positive impacts on the climate resulting from the energy production through the use of a non-polluting renewable source, to the detriment of the burning fossil fuels, thus contributing to the minimization of CO2 emissions equivalent.	High	Positive
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